

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

20/20

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

192511445

I Shank Prakash Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Shank Prakash Reddy

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly

Criteria	Weightage (%)	Level 4 (Exemplary) reasoning	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

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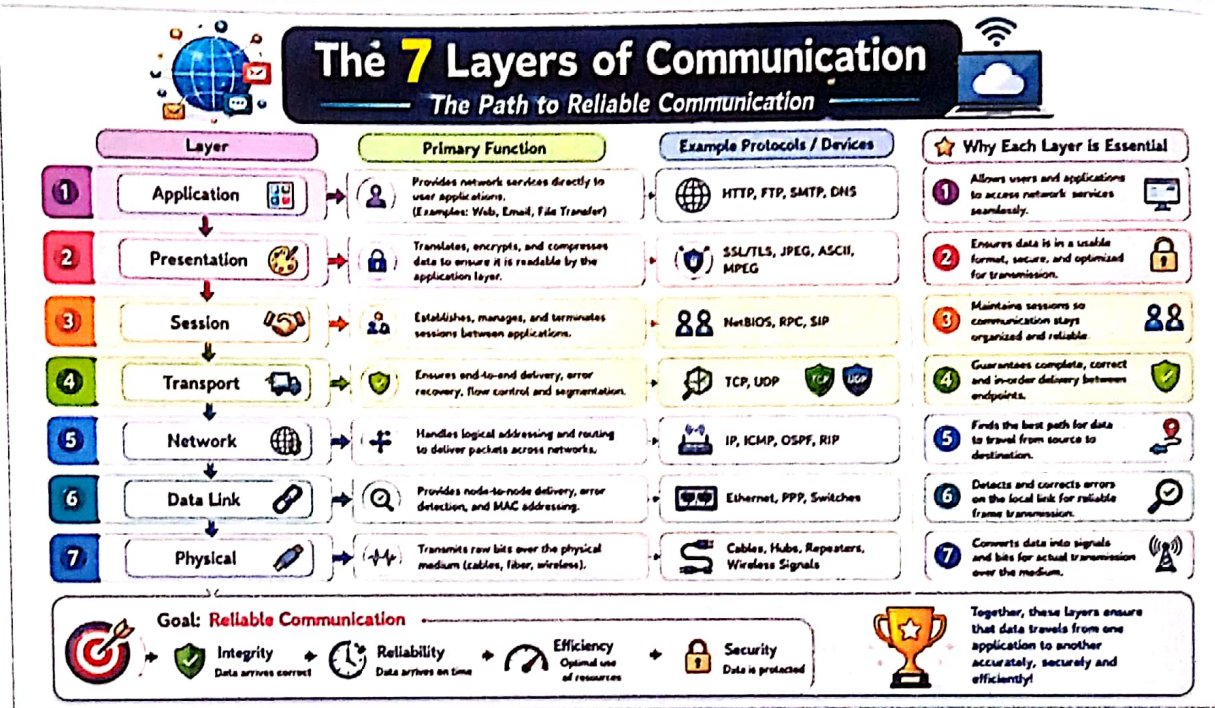
SLOT: B

Comprehensive Analysis and Concept Mapping in Computer Networks

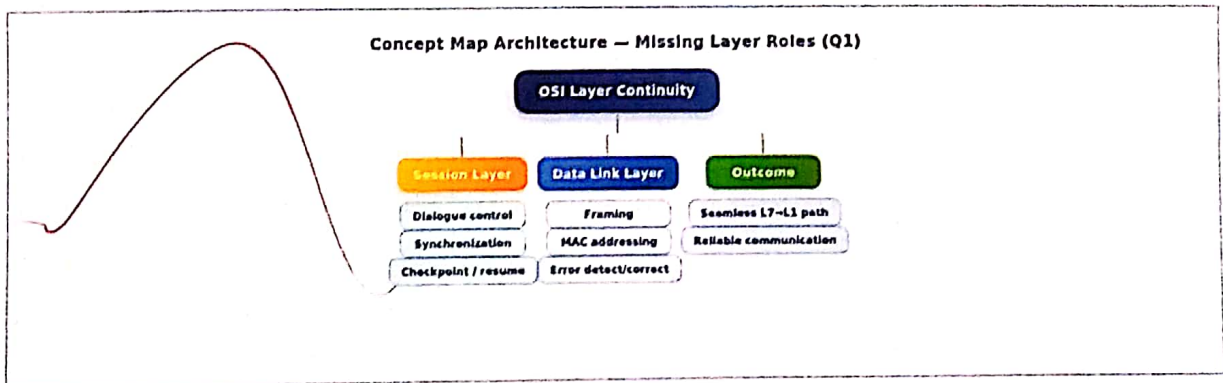
Q1. Analysing the Missing OSI Layers in the Communication Chain

The Open Systems Interconnection (OSI) model divides network communication into seven cooperating layers, and the sequence Application → Presentation → ? → Transport → Network → ? → Physical highlights two critical gaps that must be filled with the Session and Data Link layers respectively. The Session layer sits between Presentation and Transport and is responsible for establishing, managing, and gracefully terminating the dialogue between two communicating end systems; it introduces synchronization points (checkpoints) so an interrupted exchange can resume without restarting from scratch, which is vital for long-lived or resource-intensive sessions. The Data Link layer sits between Network and Physical and governs node-to-node delivery across a single physical link: it frames raw bits into structured units, attaches source and destination MAC addresses, and performs error detection so that only clean frames are passed upward. Without the Session layer, applications would lose all dialogue state on any interruption; without the Data Link layer, the Network layer would receive unstructured, unverified bit streams. Restoring both layers therefore closes the functional gap between application-level communication and physical-level bit transmission, and is essential to guaranteeing reliable, ordered, and synchronized data transmission across a network.

Flow Chart (Q1)



Concept Map Architecture (Q1)



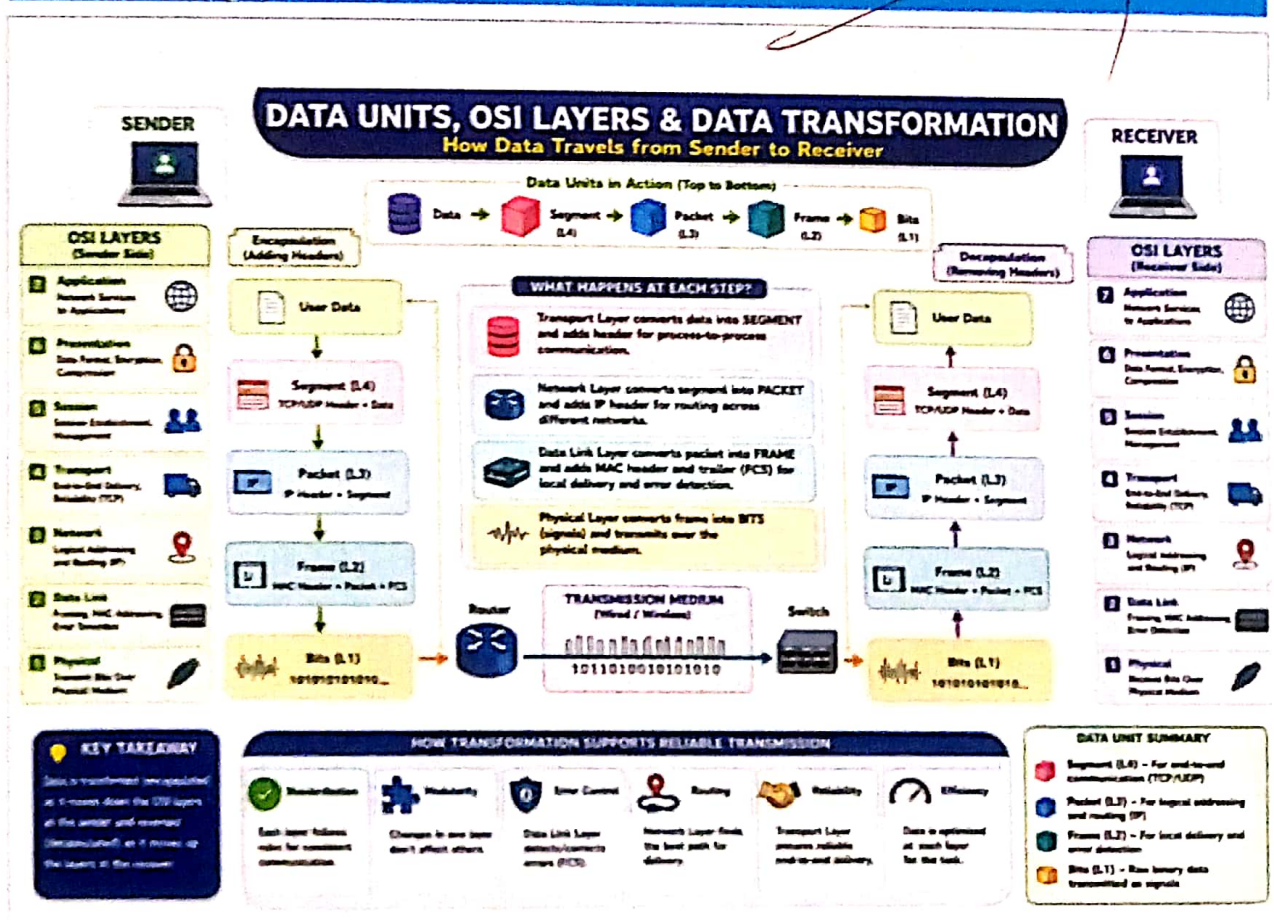
Literature Insights & Evidence (Q1)

- **Tanenbaum & Wetherall (2021):** Present the OSI reference model and describe the specific responsibilities of each of its seven layers.
- **Kurose & Ross (2022):** Explain layered network architecture and the service each layer provides to the layer above it.
- **Forouzan (2017):** Details the functions of the Session and Data Link layers within the OSI framework.
- **Peterson & Davie (2021):** Discuss layering as a design principle for building reliable network systems.
- **Stallings (2020):** Describes protocol architecture and the role of session management and framing in reliable transmission.

Q2. Linking Data Units with OSI Layers and Data Transformation

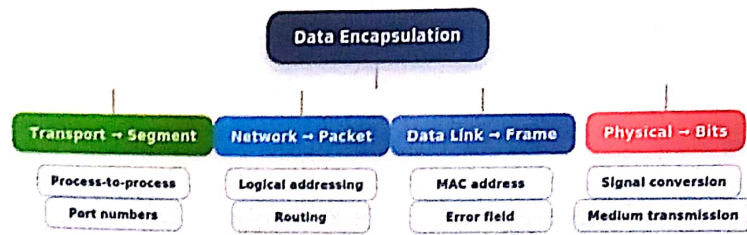
As user data descends the OSI stack for transmission, it is progressively encapsulated, and its designation changes at every layer to reflect the control information added. At the Transport layer, application data is broken into Segments, each tagged with source and destination port numbers so the receiving host can direct data to the correct process. At the Network layer, each Transport segment is wrapped into a Packet, headered with source and destination IP addresses that enable routers to make forwarding decisions across interconnected networks. At the Data Link layer, packets are further wrapped into Frames, which carry MAC addresses for identifying devices on the same physical segment along with a trailer field used for error checking. Finally, the Physical layer converts frames into a raw stream of Bits, represented as electrical voltages, light pulses, or radio waves suited to the transmission medium. This layered transformation ensures that each layer contributes precisely the addressing and control information it is responsible for, and the reverse process (decapsulation) at the receiver strips these headers in order, enabling accurate reconstruction of the original data and supporting dependable, structured transmission.

Flow Chart (Q2)



Concept Map Architecture (Q2)

Concept Map Architecture — Encapsulation Hierarchy (Q2)



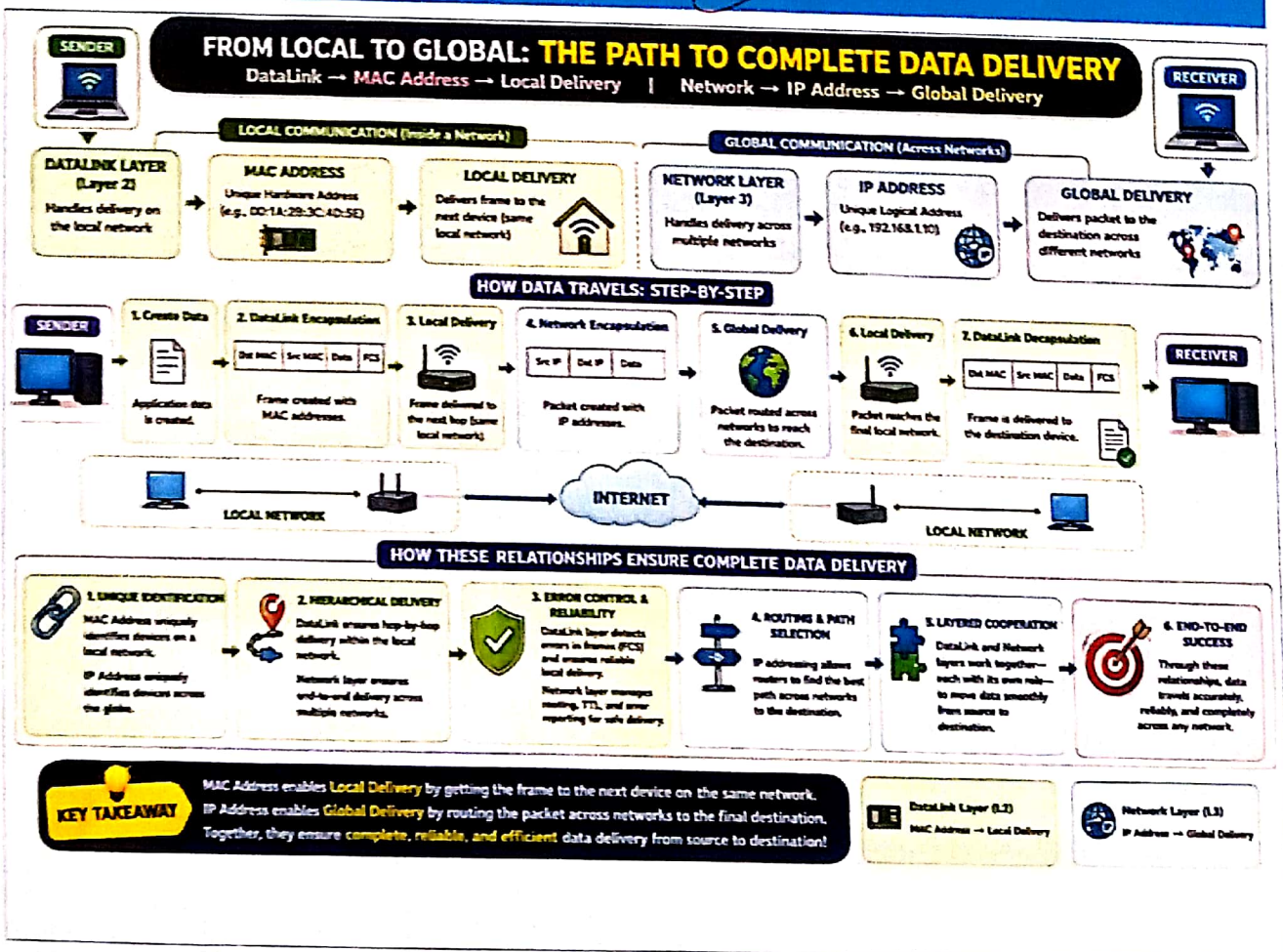
Literature Insights & Evidence (Q2)

- **Kurose & Ross (2022):** Describe the encapsulation process and the naming convention for data units at each layer.
- **Tanenbaum & Wetherall (2021):** Explain how segments, packets, frames, and bits relate to specific OSI layers.
- **Forouzan (2017):** Discusses header addition and the transformation of data units during transmission.
- **Peterson & Davie (2021):** Covers the systems-level view of encapsulation and decapsulation.
- **Comer (2018):** Explains TCP/IP data unit terminology and layer-specific headers.

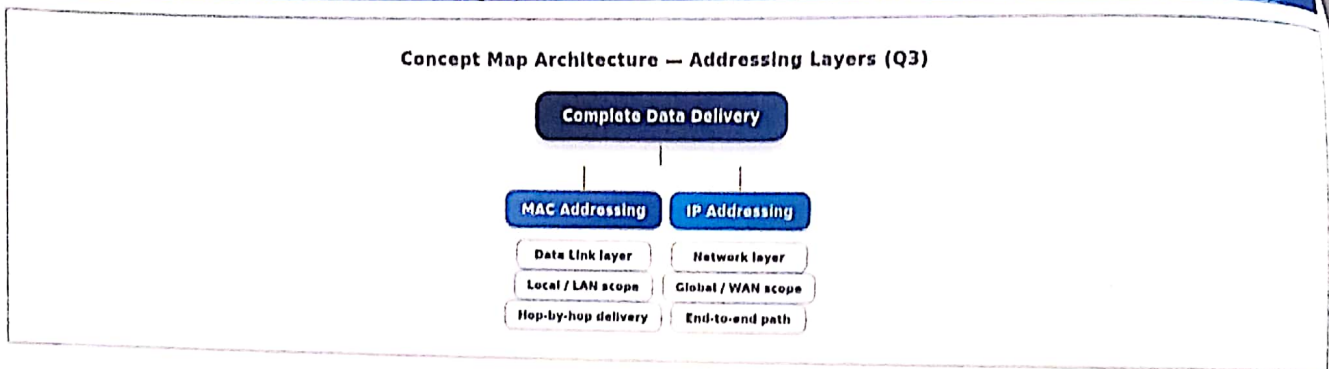
Q3. MAC Address and IP Address Relationships in Complete Data Delivery

Complete data delivery across a network relies on two complementary addressing schemes operating at different layers. The Data Link layer uses the MAC (Media Access Control) address, a fixed, hardware-burned identifier that uniquely distinguishes a network interface within a local network segment; frames are switched using MAC addresses, enabling local delivery to a device on the same LAN. The Network layer uses the IP address, a logical and hierarchical identifier organized around network topology; routers examine destination IP addresses to forward packets across multiple interconnected networks, enabling global delivery to any reachable host on the internet. These two addressing mechanisms work in tandem: IP addressing determines the overall end-to-end path a packet must travel across potentially many networks, while MAC addressing governs the specific hop-by-hop delivery within each individual network segment along that path, resolved dynamically at each hop through protocols such as ARP. Together, MAC-based local delivery and IP-based global delivery form a continuous addressing chain that ensures data reaches its correct destination completely and accurately, regardless of how many intermediate networks it must traverse.

Flow Chart (Q3)



Concept Map Architecture (Q3)



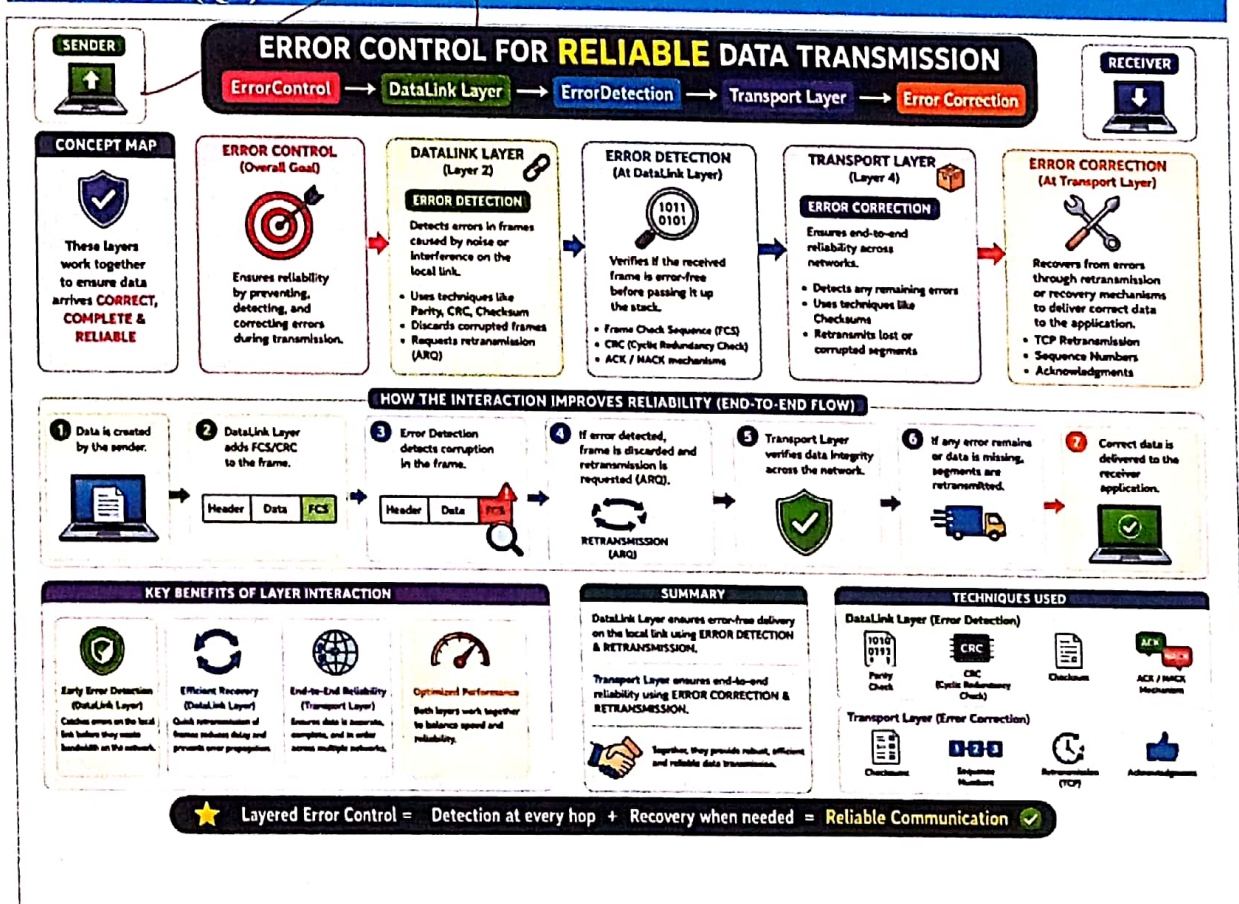
Literature Insights & Evidence (Q3)

- **Tanenbaum & Wetherall (2021):** Distinguish between physical (MAC) addressing and logical (IP) addressing.
- **Kurose & Ross (2022):** Explain how routers use IP addresses for forwarding across networks.
- **Forouzan (2017):** Details the role of ARP in resolving IP addresses to MAC addresses.
- **Comer (2018):** Discusses the structure and scope of IP addressing across interconnected networks.
- **Stallings (2020):** Covers local versus wide-area addressing schemes in network architectures.

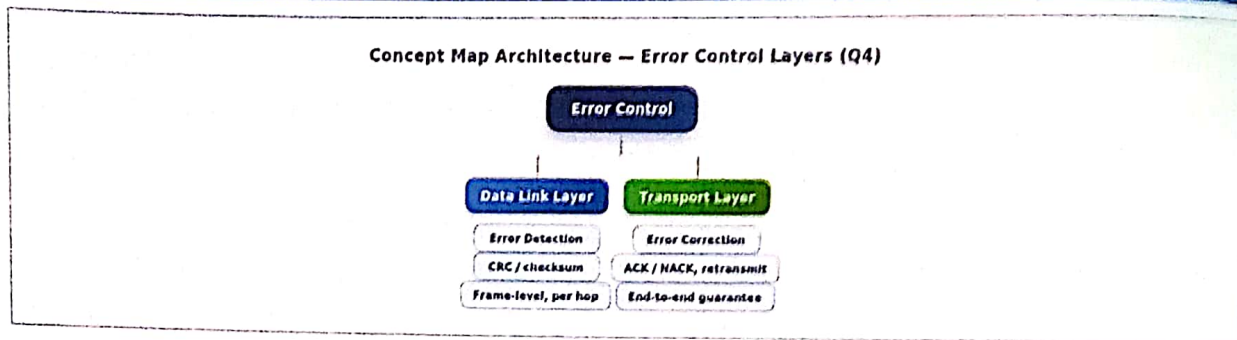
Q4. Interaction Between Data Link and Transport Layer Error Control

Reliable data transmission depends on error control mechanisms distributed across multiple OSI layers rather than concentrated at a single point, creating a layered defence against transmission faults. At the Data Link layer, error detection is performed on a per-frame, per-hop basis using techniques such as checksums or Cyclic Redundancy Check (CRC); this catches bit errors introduced as a frame crosses a single physical link between two directly connected nodes, allowing corrupted frames to be discarded before they propagate further. At the Transport layer, error correction is performed end-to-end using mechanisms such as sequence numbers, acknowledgements (ACK/NACK), and retransmission timers; this guarantees that data ultimately arrives correctly and completely at the destination application, even if a lower-layer check missed an error or a segment was lost entirely somewhere along a multi-hop path. The interaction between these two layers is complementary: Data Link detection provides fast, local correctness checks that reduce unnecessary retransmissions, while Transport correction provides the end-to-end guarantee that no single link-level check can offer on its own. This layered division of responsibility significantly improves the overall reliability of data transmission across complex, multi-hop networks.

Flow Chart (Q4)



Concept Map Architecture (Q4)



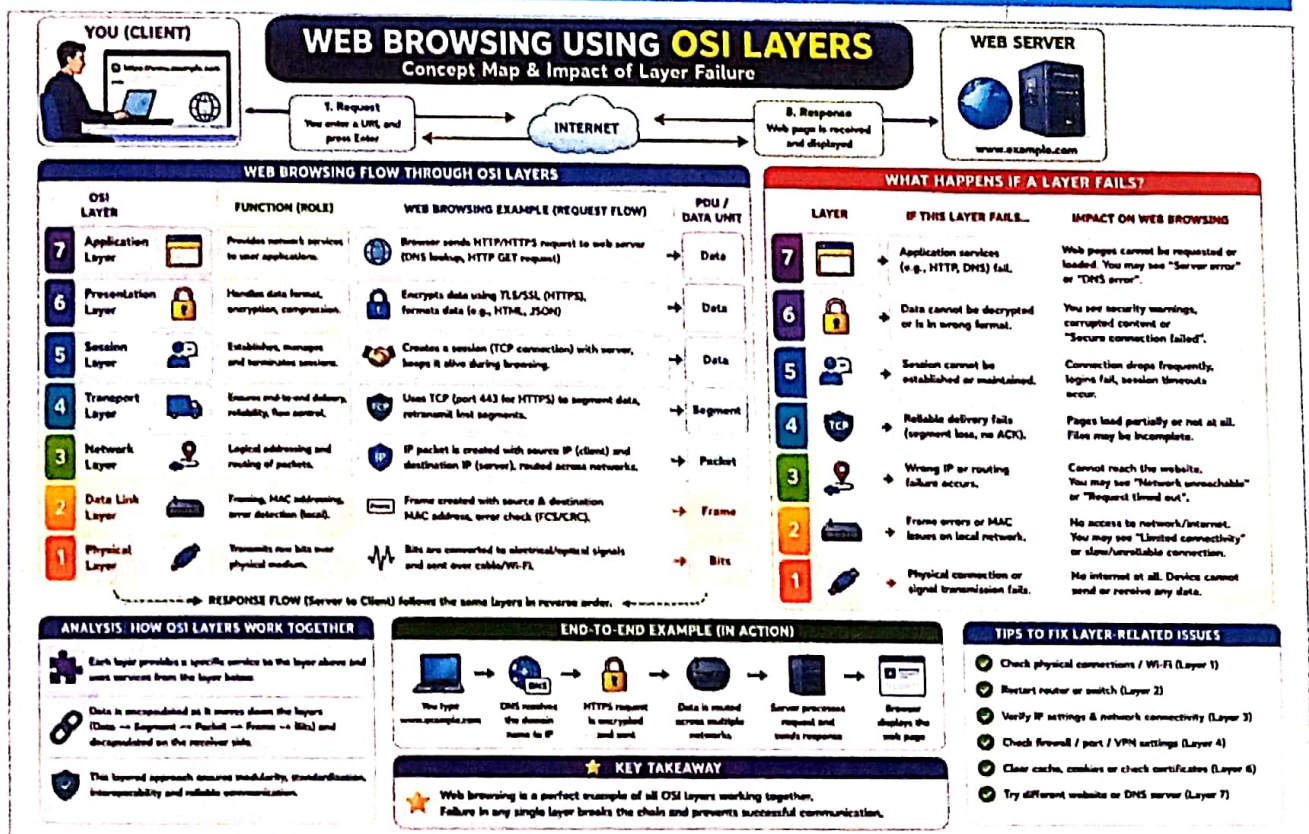
Literature Insights & Evidence (Q4)

- **Forouzan (2017):** Explains error detection techniques including checksums and CRC at the Data Link layer.
- **Kurose & Ross (2022):** Describe reliable data transfer principles used by the Transport layer.
- **Tanenbaum & Wetherall (2021):** Discuss sliding window protocols and retransmission-based error correction.
- **Peterson & Davie (2021):** Cover layered reliability mechanisms across the protocol stack.
- **Stallings (2020):** Details ARQ (Automatic Repeat reQuest) techniques for end-to-end error correction.

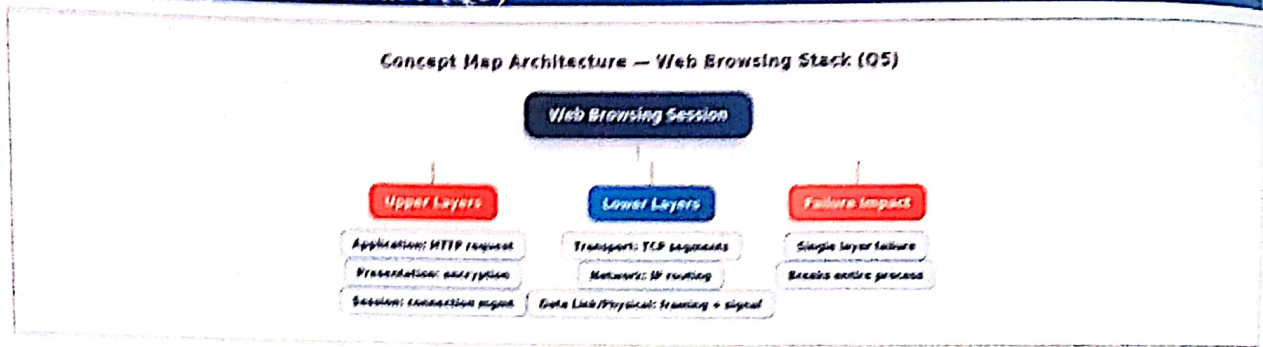
Q5. Concept Map for Web Browsing Across OSI Layers

Everyday web browsing is a practical demonstration of the full OSI stack operating in concert. When a user requests a webpage, the Application layer generates an HTTP or HTTPS request via the browser; the Presentation layer encrypts and formats this data, typically using TLS/SSL; the Session layer establishes and maintains the logical connection session with the web server for the duration of the exchange. The Transport layer, generally using TCP, segments this data and provides reliable, ordered delivery through acknowledgements and retransmission. The Network layer routes these segments as IP packets across the interconnected networks of the internet toward the server's IP address. The Data Link layer frames the data for transmission across each individual physical hop using MAC addressing, and the Physical layer converts these frames into electrical, optical, or radio signals for actual transmission over the medium. Because each layer relies entirely on the services provided by the layer beneath it, a failure at any single layer disrupts the entire communication process: a severed cable at the Physical layer halts all signal transmission; a routing misconfiguration at the Network layer prevents packets from reaching the server; and a crashed web server at the Application layer means no valid response is ever generated, even if every lower layer functions perfectly. Reliable web browsing therefore depends on the correct, coordinated functioning of all seven layers simultaneously.

Flow Chart (Q5)



Concept Map Architecture (Q5)



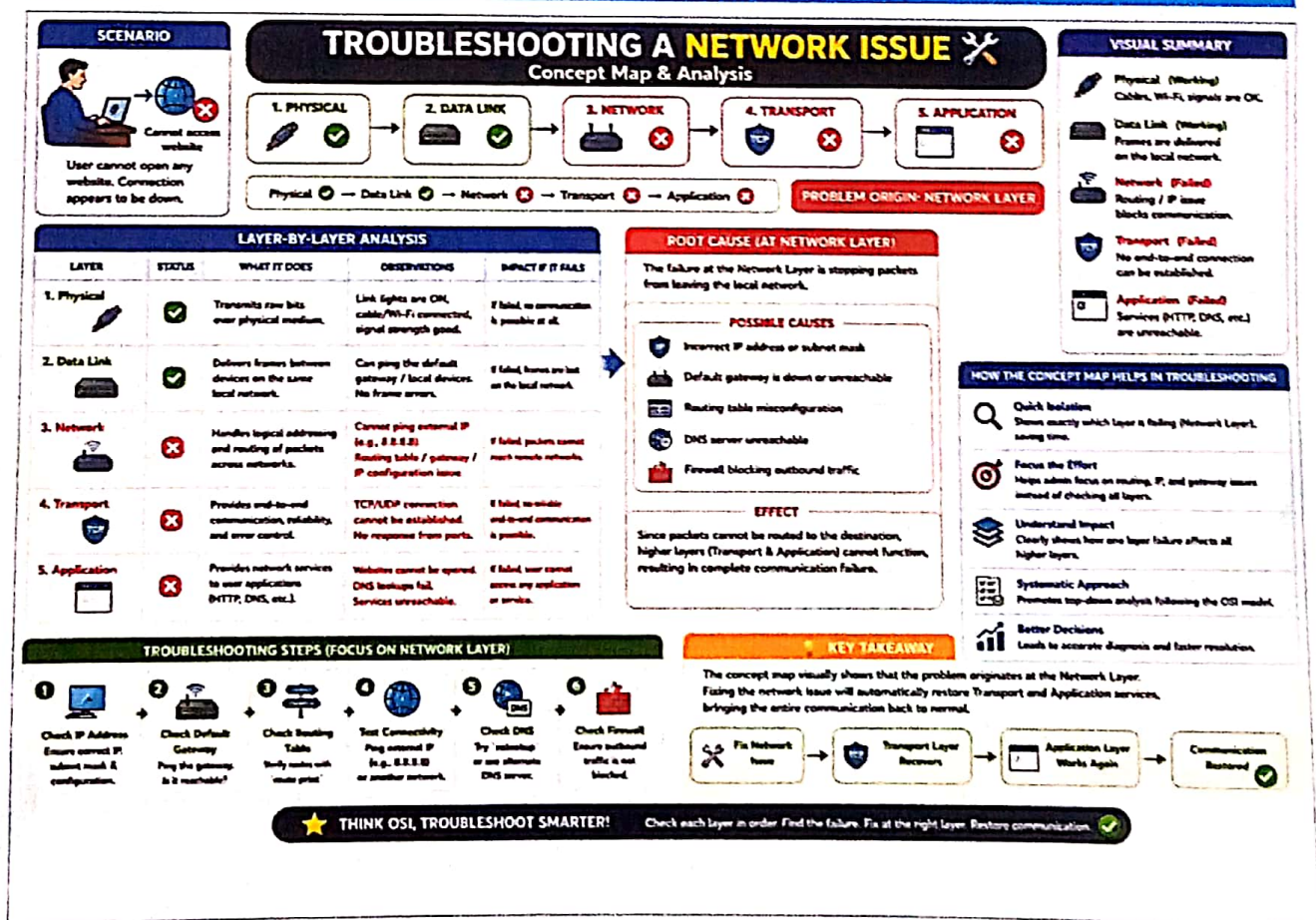
Literature Insights & Evidence (Q5)

- Kurose & Ross (2022): Use web browsing (HTTP over TCP/IP) as a central example of layered protocol interaction.
- Tanenbaum & Wetherall (2021): Explain how TLS, TCP, and IP interact to support secure web communication.
- Forouzan (2017): Describe the role of each OSI layer in a complete client-server data exchange.
- Peterson & Davie (2021): Discuss the end-to-end argument and layer dependency in networked applications.
- Comer (2018): Cover the TCP/IP protocol suite as applied to internet-based application delivery.

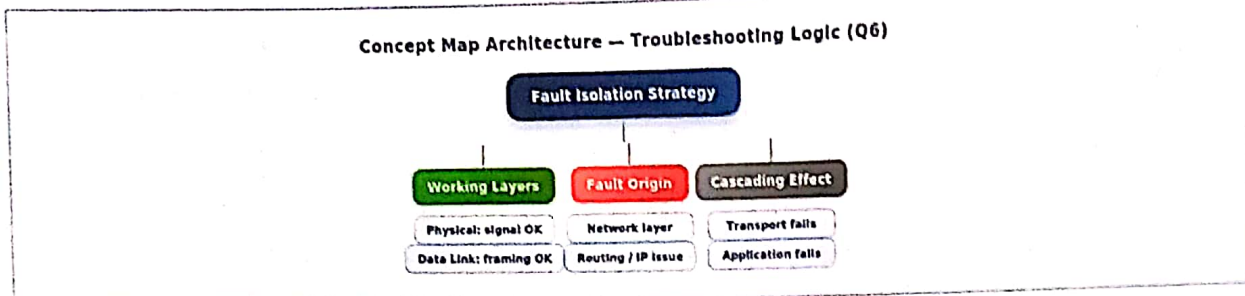
Q6. Troubleshooting a Network Issue Using a Concept Map

Concept maps that annotate each OSI layer with a pass/fail status are a powerful diagnostic tool for systematic network troubleshooting. In the scenario presented, the Physical and Data Link layers are both functioning correctly, confirmed by successful signal transmission and successful local frame delivery, but the Network layer is marked as failing, with the failure then propagating upward to also affect the Transport and Application layers. Because every layer above the Network layer depends on its services to route data correctly, a single fault at the Network layer, such as an incorrect routing table entry, a misconfigured default gateway, or an IP addressing conflict, is sufficient to prevent Transport-layer segments from being delivered and, consequently, prevent any usable data from reaching the Application layer, even though neither of those upper layers is actually the root cause of the problem. This concept map supports efficient troubleshooting by encouraging a bottom-up diagnostic approach: because Physical and Data Link have already been confirmed operational, a network engineer can skip re-checking cabling or switches and instead focus diagnostic effort directly on Network-layer configuration, rather than wasting time investigating the Application layer, where the symptoms appear but the fault does not originate.

Flow Chart (Q6)



Concept Map Architecture (Q6)



Literature Insights & Evidence (Q6)

- **Stallings (2020):** Discuss systematic fault diagnosis using layered network models.
- **Tanenbaum & Wetherall (2021):** Explain how layer dependency causes upper-layer symptoms from lower-layer faults.
- **Kurose & Ross (2022):** Describe common causes of network layer failures, including routing misconfiguration.
- **Forouzan (2017):** Cover diagnostic approaches for isolating faults within the OSI model.
- **Peterson & Davie (2021):** Discuss network management and troubleshooting strategies in layered systems.

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